Loss functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable paired-loss and comparison record.

Calculate MSE,repair denominator/alignment errors and state the objective-dependent conclusion.

Keep one habit: W1–W3/W5 → selected new comparison → project paired loss ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A loss assigns a numerical penalty to a **prediction** compared with a **target**. For this regression lesson, **residual** e = **prediction** - **target**, and **squared error** is e squared.

  Squaring makes both positive and negative **residual**s contribute nonnegative penalties. We use ordinary **mean squared error** without a one-half factor.

- The **mean squared error**, or MSE, is (1 + 1 + 4) / 3 = 2. It summarises these three examples, not every possible future **input**. If **target**s use cm, **residual**s use cm and MSE uses square cm.

Parameters and architecture

My remembered idea, example and correction

Composing layers

My remembered idea, example and correction

Dot products and weighted sums

My remembered idea, example and correction

Lesson reference for workbook practice

Example · Target · Prediction · Residual · Squared error

Example: A

Target: 1

Prediction: 2

Residual: 1

Squared error: 1

Example: B

Target: 3

Prediction: 2

Residual: -1

Squared error: 1

Example: C

Target: 2

Prediction: 4

Residual: 2

Squared error: 4

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use residual prediction–target, ordinary MSEmean of squared residuals, no half factor. Main targets[1, 3, 2]:P[2, 2, 4]res[1, –1, 2], squares[1, 1, 4], MSE 2; Q[1, 2, 2]MSE 1/3; fixed supplied baseline[2, 2, 2]MSE 2/3(no fitting provenance supplied); R[0, 3, 3]MSE 2/3. W2 targets[0, 4]/pred[2, 2]mean residual 0 butMSE4. Denominator includes all cases, even zero errors; 1/0/9 squares over 3 give10/3.

  Equal lengths cannot verify case alignment. Separate picturetargets[1, 3], A[0, 4]MSE 1, B[1, 5]MSE 2, bothMAE1. Training/selection losses are not untouched final evidence. Independent inputs are different.

- For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use residual prediction–target, ordinary MSEmean of squared residuals, no half factor. Main targets[1, 3, 2]:P[2, 2, 4]res[1, –1, 2], squares[1, 1, 4], MSE 2; Q[1, 2, 2]MSE 1/3; fixed supplied baseline[2, 2, 2]MSE 2/3(no fitting provenance supplied); R[0, 3, 3]MSE 2/3. W2 targets[0, 4]/pred[2, 2]mean residual 0 butMSE4. Denominator includes all cases, even zero errors; 1/0/9 squares over 3 give10/3.

  Equal lengths cannot verify case alignment. Separate picturetargets[1, 3], A[0, 4]MSE 1, B[1, 5]MSE 2, bothMAE1. Training/selection losses are not untouched final evidence. Independent inputs are different.

- **Target**s [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use residual prediction–target, ordinary MSEmean of squared residuals, no half factor. Main targets[1, 3, 2]:P[2, 2, 4]res[1, –1, 2], squares[1, 1, 4], MSE 2; Q[1, 2, 2]MSE 1/3; fixed supplied baseline[2, 2, 2]MSE 2/3(no fitting provenance supplied); R[0, 3, 3]MSE 2/3. W2 targets[0, 4]/pred[2, 2]mean residual 0 butMSE4. Denominator includes all cases, even zero errors; 1/0/9 squares over 3 give10/3.

  Equal lengths cannot verify case alignment. Separate picturetargets[1, 3], A[0, 4]MSE 1, B[1, 5]MSE 2, bothMAE1. Training/selection losses are not untouched final evidence. Independent inputs are different.

- A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use residual prediction–target, ordinary MSEmean of squared residuals, no half factor. Main targets[1, 3, 2]:P[2, 2, 4]res[1, –1, 2], squares[1, 1, 4], MSE 2; Q[1, 2, 2]MSE 1/3; fixed supplied baseline[2, 2, 2]MSE 2/3(no fitting provenance supplied); R[0, 3, 3]MSE 2/3. W2 targets[0, 4]/pred[2, 2]mean residual 0 butMSE4. Denominator includes all cases, even zero errors; 1/0/9 squares over 3 give10/3.

  Equal lengths cannot verify case alignment. Separate picturetargets[1, 3], A[0, 4]MSE 1, B[1, 5]MSE 2, bothMAE1. Training/selection losses are not untouched final evidence. Independent inputs are different.

- A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Target**s are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

  State what changes in the conclusion.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Compare an unfamiliar evaluation independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled loss functions **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate per-example **squared error** and **mean squared error**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Expose cancellation for Loss functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Loss functions. Proposed explanation: "Average **residual** is the loss". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Target**s are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

  State what changes in the conclusion.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

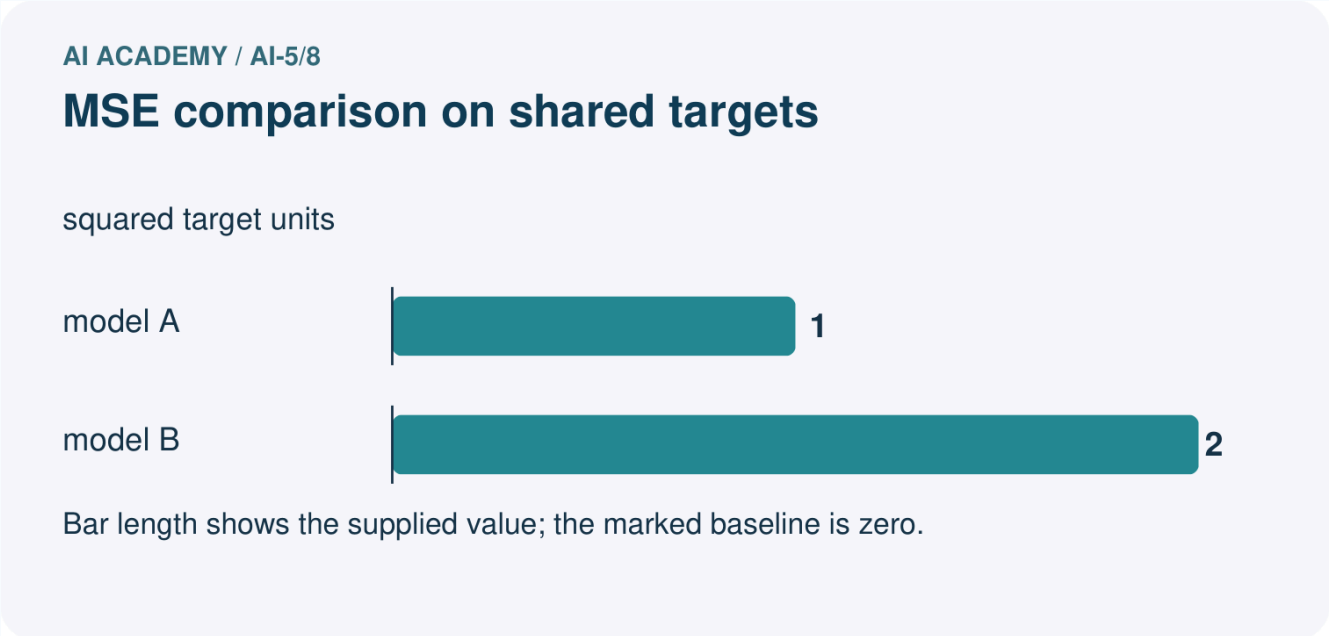
- Compare candidate losses with the same **target**s and denominator.
- **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/8

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A squared errors

A residuals [-1, 1] give squares [1, 1], MSE 1.

↓

2. B squared errors

B residuals [0, 2] give squares [0, 4], MSE 2.

↓

3. Compare objective

A wins under MSE; both MAEs equal 1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare candidate losses with the same **target**s and denominator.
- Which labelled step matches this detail? A wins under MSE; both MAEs equal 1.
- 1. Compare objective
- 2. A **squared error**s
- 3. B **squared error**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> ---
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> ---
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare objective A wins under MSE; both MAEs equal 1. For this loss calculation, the signed error is **prediction** minus **target**.

MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more.

State the objective before selecting a model, because these two objectives give different comparisons.

Changed-case question and optional hint

- Changed-case practice: Choose MAE instead of MSE as the sole criterion. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. A and B tie at MAE 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A and B tie at MAE 1. **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Repair and retain the record

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Loss functions: the case

AI ACADEMY / AI-5/8 · 01 · LOOK AT THE FACTS

Loss functions: the case

The situation

Compare candidate losses with the same targets and denominator.

Supplied evidence

Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain loss functions using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

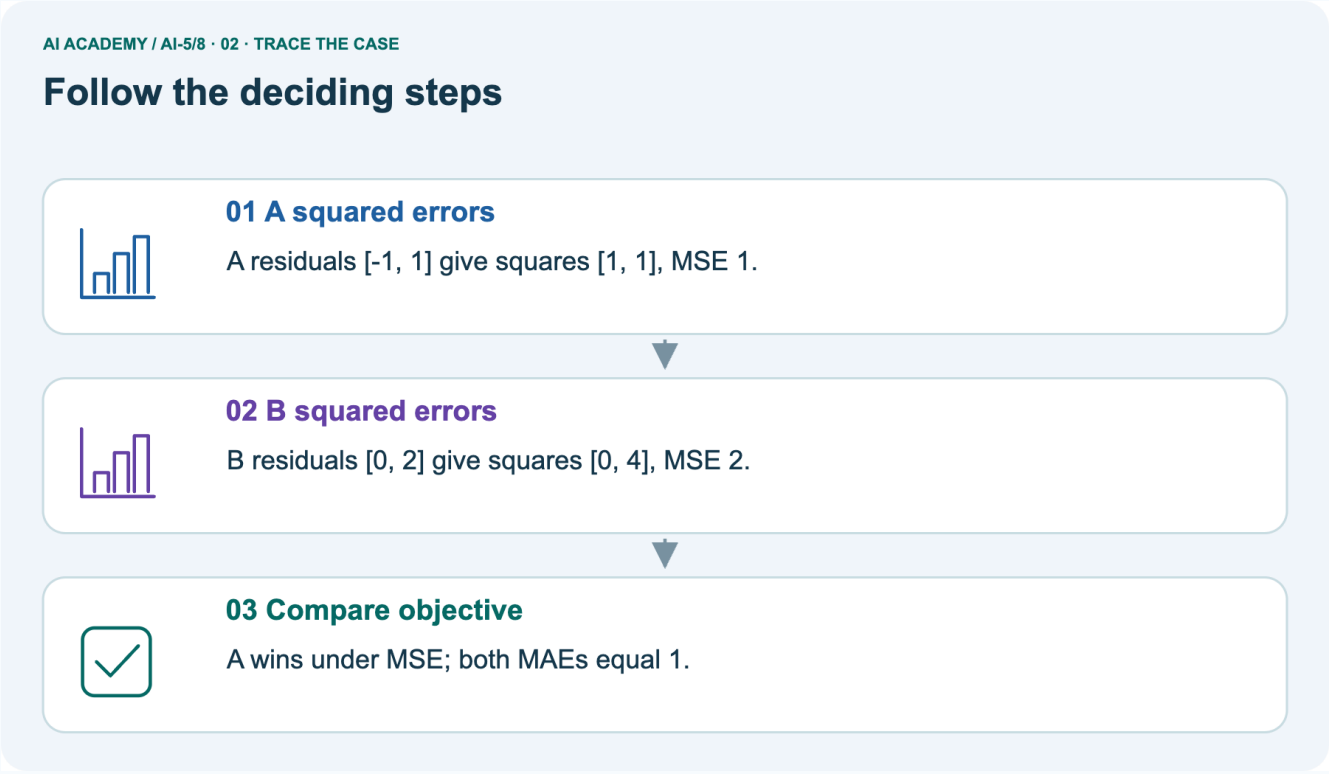
Compare candidate losses with the same **target**s and denominator. Supplied facts: **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



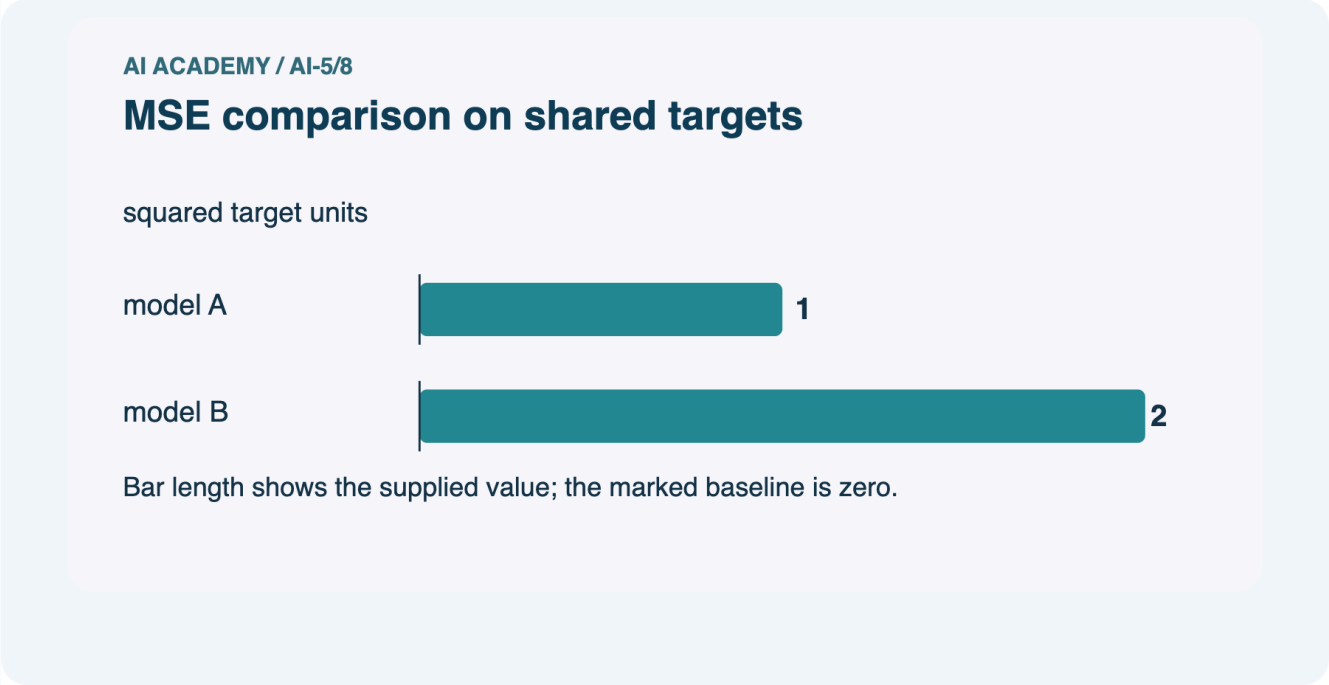
Follow the labelled picture with the matching explanation. Open the vector diagram

A **squared error**s: A **residual**s [-1, 1] give squares [1, 1], MSE 1. → B **squared error**s: B **residual**s [0, 2] give squares [0, 4], MSE 2. → Compare objective: A wins under MSE; both MAEs equal 1.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. MSE comparison on shared targets



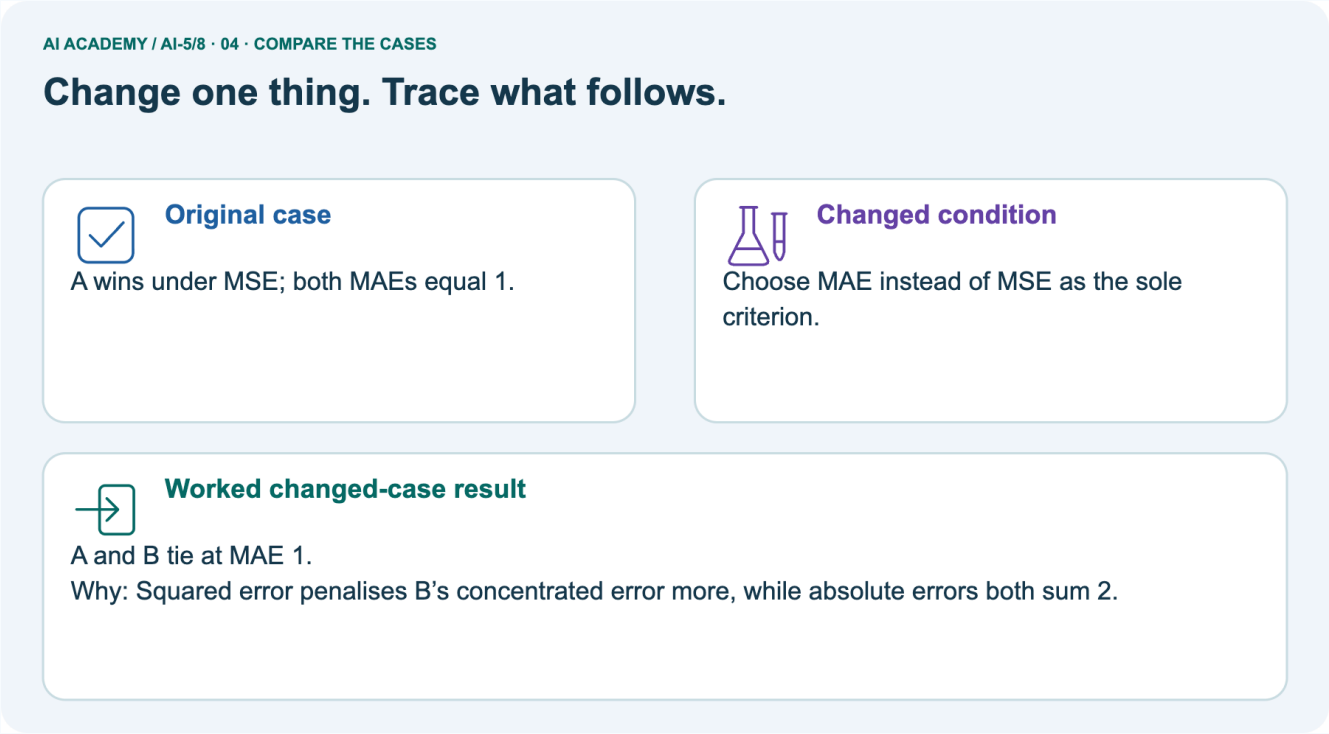
Follow the labelled picture with the matching explanation. Open the vector diagram

MSE comparison on shared **target**s. Units: squared **target** units. model A: 1. model B: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

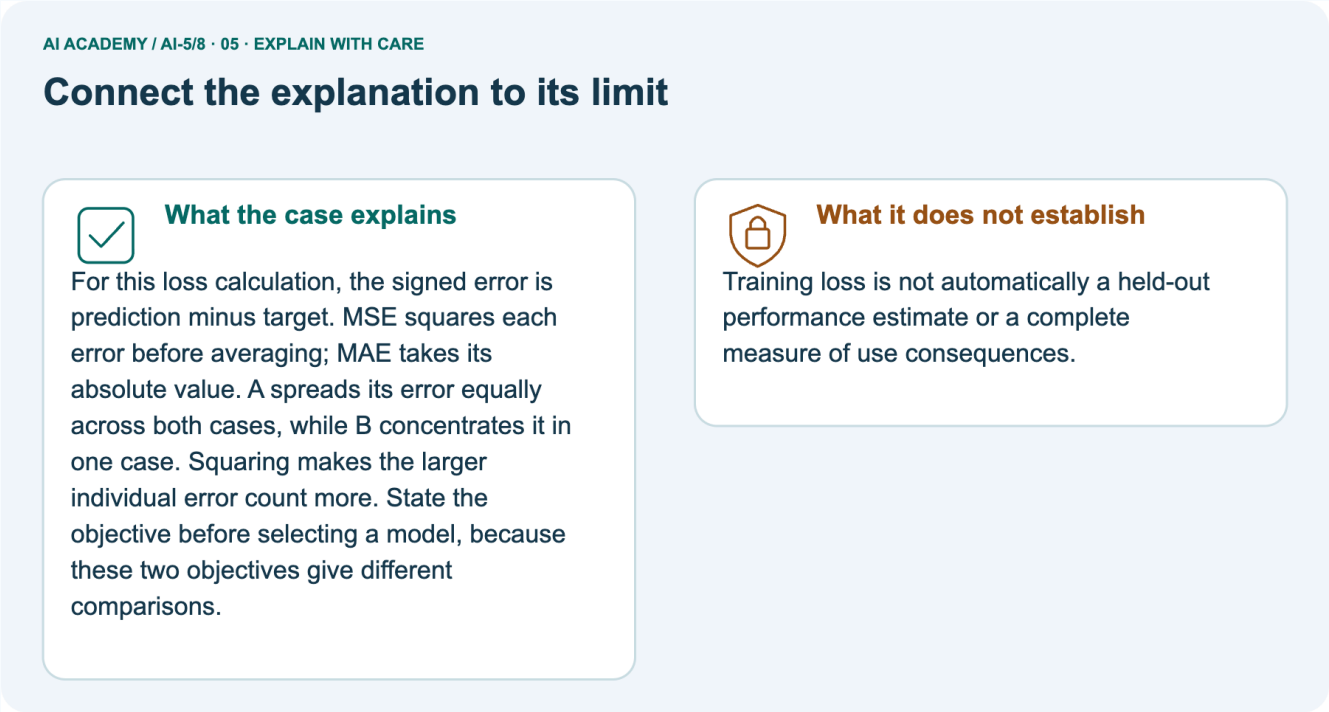
Original case: A wins under MSE; both MAEs equal 1. Changed condition: Choose MAE instead of MSE as the sole criterion. Expected result: A and B tie at MAE 1.

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

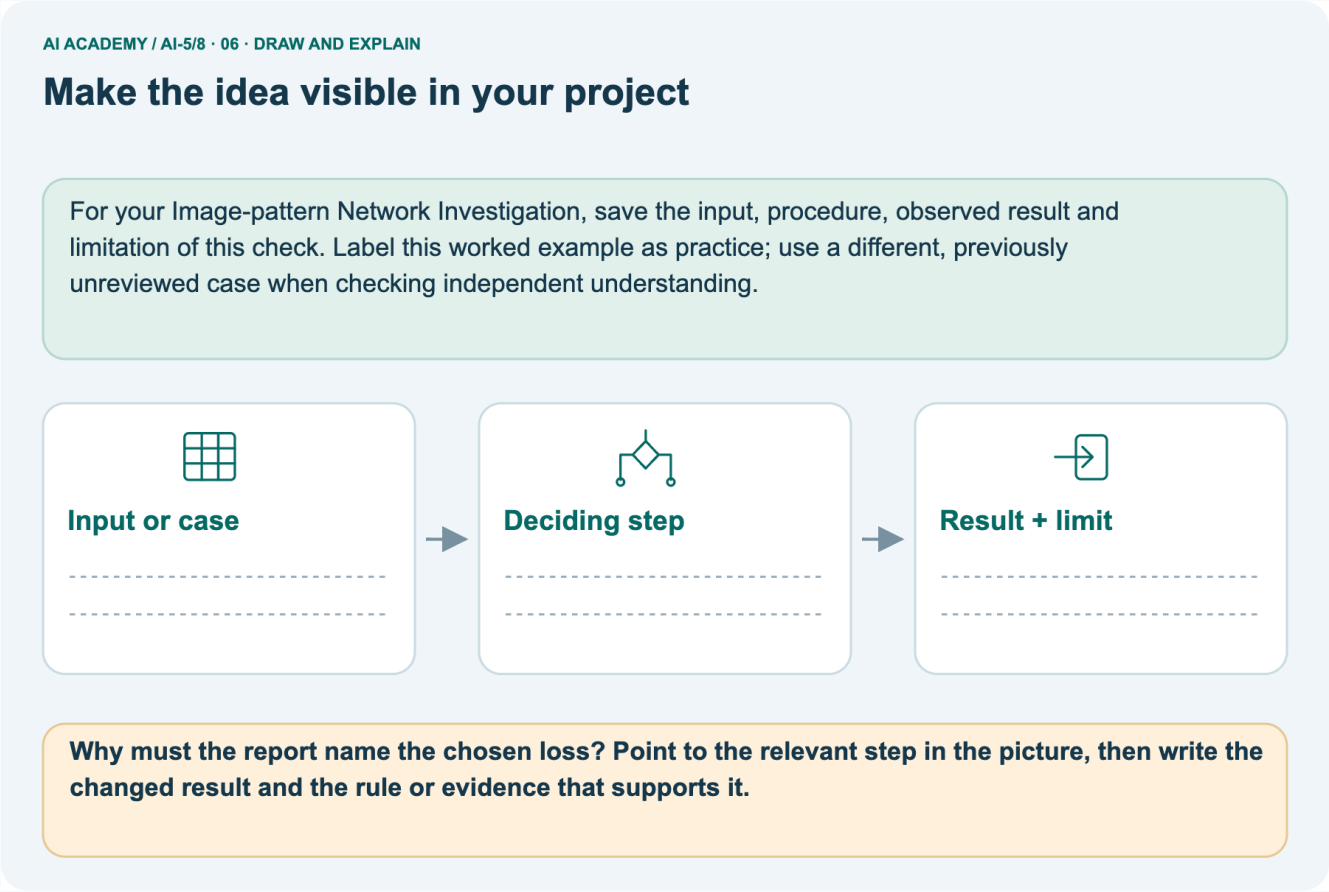
Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Limit: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Loss functions: the case

02. Follow the deciding steps

03. MSE comparison on shared targets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.